MA378: Tutorial Sheet 4 ANS with some solutions

Q1. Deduce the 4-point Newton-Cotes Rule for estimating the integral $\int_0^1 f(x)dx$:

$$Q_3(f) = q_0 f(x_0) + q_1 f(x_1) + q_2 f(x_2) + q_3 f(x_3).$$

Extend the rule to estimate the integral of functions over $[a, b]$.

Q2. Prove the error bound given for the Trapezium rule. That is, show that

$$|\int_a^b f(x)dx - Q_1(f)| := \mathcal{E}_1 \leq \frac{(b-a)^3}{12} M_2.$$

Q3. Explain clearly, with an example, why in general it is not true that $Q_n(f) \to \int_a^b f(x)dx$ as $n \to \infty$.

> **Answer:** Runge's example again...

Q4. (i) Deduce an error estimate for the Composite Trapezium Rule.

(ii) Taking $N = 10$, give an upper bound for the error in the Composite Trapezium Rule when approximating $\int_1^2 \ln(x)dx$.

(iii) What value of N would you have to take to ensure that the error was less that 10^{-5}?

Q5. (i) Deduce the formula for the *composite Simpson's Rule*.

(ii) Derive an error estimate for the *composite Simpson's Rule*.

(iii) What value of N would you have to take to ensure that the error in the estimate of $\int_1^2 \ln(x)dx$ is less that 10^{-6}?

(iv) Denote the $(N+1)$-point Composite Simpson's Rule by $S_N(f) \approx \int_a^b f(x)dx$. Show that, for sufficiently smooth $f(x)$,

$$\lim_{n\to\infty} S_N(f) = \int_a^b f(x)dx.$$

Q6. Determine the precision of the following schemes for estimating $\int_0^1 f(x)dx$.

(i) $Q(f) = f(\frac{1}{2})$. [Note: closely related to a question on Assignment 2, so won't be done in tutorials]

> **Answer:** $Q(1) = 1 = \int_0^1 1dx$ and $Q(x) = 1/2 = \int_0^1 xdx$. But $Q(x^2) = 1/4 \neq \int_0^1 x^2dx$. So this method has precision 1. FYI, this is the so-called mid-point rule. It is the 1-point Gaussian Quadrature Rule.

(ii) $Q(f) = \frac{1}{4}f(0) + \frac{3}{4}f(\frac{2}{3})$.

> **Answer:** $Q(1) = 1 = \int_0^1 1dx$, $Q(x) = 1/2 = \int_0^1 xdx$, $Q(x^2) = 1/3 = \int_0^1 x^2dx$. But $Q(x^3) = 2/9 \neq \int_0^1 x^3dx$. So this method has precision 2.

(iii) $Q(f) = \frac{3}{2}f(\frac{1}{3}) - 2f(\frac{1}{2}) + \frac{3}{2}f(\frac{2}{3})$.

> **Answer:** $Q(x^k) = 1/(k+1) = \int_0^1 x^k dx$, for $k = 0, 1, 2, 3$. But $Q(x^4) = 41/216 \neq \int_0^1 x^4 dx$. So $Q(\cdot)$ has precision 3.

Q7. (Based on the 2023/24 exam) Consider the following rule for approximating $\int_0^4 f(x)dx$:

$$R(f) = q_0 f(0) + q_1 f(1) + \frac{2}{9} f(3) + f(4).$$

(a) Determine the values of q_0 and q_1 so that $R(\cdot)$ has precision 1.

> **Answer:** A method has precision 1 if it exact for all polynomials of degree 1 or less.
> If we require that it be exact for $f(x) = 1$, then we get the equation $q_0 + q_1 = 25/9$.
> If we require that it be exact for $f(x) = x$, then we get the equation $q_1 = 10/3$. From this we determine that $q_0 = -5/9$.

(b) What is the maximum precision of $R(\cdot)$ for the values of q_0 and q_1 that you have determined?

> **Answer:** First note that, since both the integral operator and $R(\cdot)$ are linear, if $R(\cdot)$ is correct for $1, x, x^2, \ldots, x^n$ it is exact for all polynomials of degree n or less.
> We already know that $R(\cdot)$ has precision of t least 1.
> Next we check it for $f(x) = x^2$. For that we get $R(x^2) = 64/3 = \int_0^4 x^2$, so it has precision at least 2.
> Then we check it for $f(x) = x^3$. For that we get $R(x^3) = 220/3 \neq 64 = \int_0^4 x^2$, so it does not have precision 3.

(c) Why is $R(\cdot)$ not a Newton-Cotes rule, in the strictest sense?

> **Answer:** N-C methods use equally spaced quadrature points, which is not the case here.

Q8. Use a change of variables, as we did with the Trapezium rule, to show that the rule for approximating $\int_0^1 f(x)dx$ is

$$G_1(f) = \frac{1}{2}\Big(f\big(\frac{1}{2} - \frac{1}{2\sqrt{3}}\big) + f\big(\frac{1}{2} + \frac{1}{2\sqrt{3}}\big) \Big).$$

More generally, extend the rule to an arbitrary interval $[a, b]$.

Q9. Use $G_1(x)$ to estimate $\displaystyle\int_1^2 \ln(x)dx$. How does this compare with the Trapezium and Simpson's Rule?

Q10. Derive a 3-point Gaussian Quadrature Rule to estimate $\int_{-1}^1 f(x)dx$. *Hint:* $x_1 = 0$.

Answer: The method is $G_2(f) := w_0 f(x_0) + w_1 f(x_1) + w_2 f(x_2)$, and, since it has 6 degrees of freedom, it should be exact for each of the polynomials in $\{1, x, x^2, x^3, x^4, x^5\}$. These 6 polynomials will lead to 6 (nonlinear) equations. However, since we know that $x_1 = 0$, we need only 5. In any case the equations are

$(i)\ \ w_0 + w_1 + w_2 = 2$	$(ii)\ \ w_0x_0 + w_2x_2 = 0$	$(iii)\ \ w_0x_0^2 + w_2x_2^2 = 2/3$
$(iv)\ \ w_0x_0^3 + w_2x_2^3 = 0$	$(v)\ \ w_0x_0^4 + w_2x_2^4 = 2/5$	$(vi)\ \ w_0x_0^5 + w_2x_2^5 = 0$

(ii) Gives that $w_0x_0 = -w_2x_2$. Substitute this into (iv) to get that $(-w_2x_2)x_0^2 - w_2x_2^3 = 0$. Since $x_2 \neq 0$, and $w_2 \neq 0$, we can deduce that $x_0^2 = x_2^2$. So $x_0 = -x_2$, because $x_0 < x_2$. Again using (ii) we get $w_0 = w_2$. Next use (iii) to see that $w_0x_0^2 = 1/3$, and (v) to give $w_0x_0^4 = 1/5$. Combining those leads to $x_02 = 3/5$. So now we have that $x_0 = -\sqrt{3/5}$ and $x_1 = \sqrt{3/5}$. Reusing $w_0x_0^2 = 1/3$, we have that $w_0 = 5/9 = w_2$. Finally, (i) gives $w_1 = 8/9$. That is, the method is

$$G_2(f) = \frac{5}{9}f\big(-\sqrt{\frac{3}{5}}\big) + \frac{8}{9}f(0) + \frac{5}{9}f\big(\sqrt{\frac{3}{5}}\big).$$